Name: ______________________

Worksheet

Inequality notation to express error

Task A: Range of values on a number line

1) Identify the **range** of unrounded values when:

a) 34 has been rounded to the nearest integer.

b) 63 has been rounded to the nearest integer.

c) 1 has been rounded to the nearest integer.

2) Identify the **range** of unrounded values when:

a) 6 has been rounded to 1 s.f.

b) 56 has been rounded to the 2 s.f.

c) 1000 has been rounded to 4 s.f.

3) Work out the question given the following:

a) [ ] has been rounded to 1 [ ] s.f.

Smallest: 3.5

Largest: $4.4\dot{9}$

b) [ ] has been rounded to the [ ] s.f.

Smallest: 97.5

Largest: [ ]

c) [ ] has been rounded to [ ]

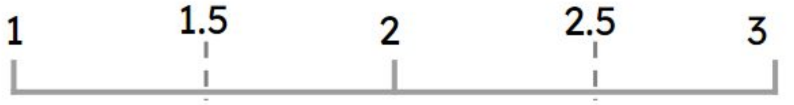
Smallest: 99.5

Largest: [ ]

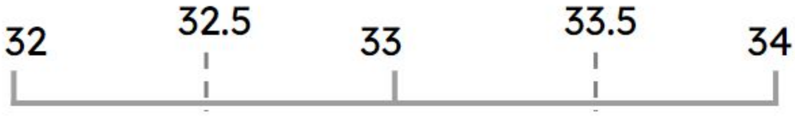
Task B: Error intervals

1) Given the number lines, identify the **error interval.**

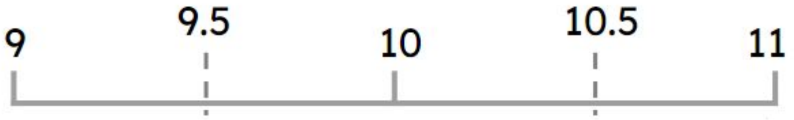
a) 2 rounded to 1 s.f.



b) 33 rounded to 2 s.f.



c) 10 rounded to 2 s.f.



2) Identify the error interval of the following:

a) 482 is rounded to the nearest whole number.

b) 65 is rounded to the 2 s.f.

c) 7.2 is rounded to 1 d.p.

d) 4.54 is rounded to 2 d.p.

3) Pair up the question with the correct **error interval**.

3 is rounded to 1 s.f.	$3.15 \leq x < 3.25$
3.3 is rounded to 1 d.p.	$3.245 \leq x < 3.255$
3.2 is rounded to 2 s.f.	$2.5 \leq x < 3.5$
3.25 is rounded to 2 d.p.	$3.25 \leq x < 3.35$